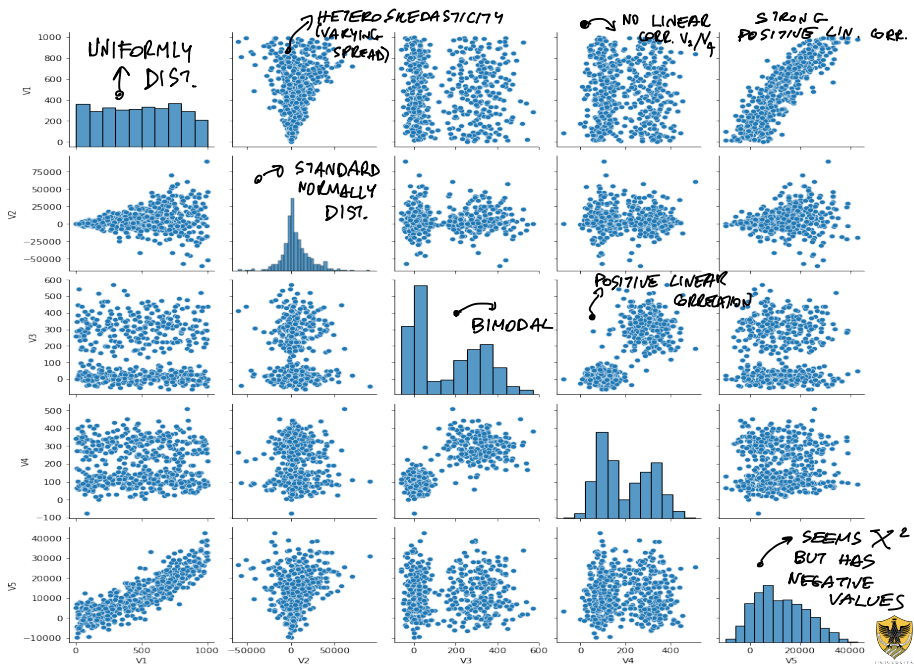




recall on correlation...

a statistical dependence (linear) between two random variables

- Pearson's correlation coefficient:

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$$

- sample version:

$$r_{XY} = \frac{\sum_{i=1, n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1, n} (x_i - \bar{x})^2} \sqrt{\sum_{i=1, n} (y_i - \bar{y})^2}}$$

DENOMINATOR
TO HAVE
CORR. $\in [-1, 1]$

- ρ_{XY} or $r_{XY} = \begin{cases} 1 & \text{positive correlation} \\ 0 & \text{linear independence} \\ -1 & \text{negative correlation} \end{cases}$

~~linear independence~~ NO CORRELATION
 $-0,3 \leq \rho \leq 0,3$ WEAK CORR.

$0,3 \leq \rho \leq 0,7$ MODERATE CORR.
($-0,7 \leq \rho \leq -0,3$)

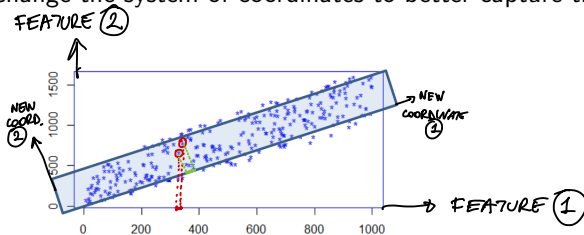
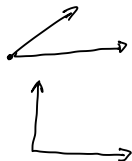
exercise 2.2

$\rho \geq 0,7 \rightarrow$ STRONG
 $\rho \leq -0,7 \rightarrow$ CORR.



beyond correlation, much **information** in data is about **distance** between points (the spread!)

- to extract information from data we have to focus on some aspects and discard the others
- data has many variables: is it possible to reduce the dimension of the data space, transform it, and preserve **most of** the distance between points? (e.g. humans like to see things in 2 dimensions)
- is it possible to change the system of coordinates to better capture the data variance?



⇒ **orthogonality** is a key aspect

recall on linear algebra

- 2 vectors are linearly dependent if one can be obtained as a linear combination of the other

$$a = \begin{pmatrix} 2 \\ 1 \\ 4 \\ 3 \end{pmatrix}, b = \begin{pmatrix} 4 \\ 2 \\ 8 \\ 6 \end{pmatrix} \Rightarrow a = 0.5b$$

- n vectors are linearly dependent if any of them can be obtained as a linear combination of the others

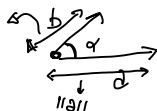
$$a = \begin{pmatrix} 2 \\ 1 \\ 4 \\ 3 \end{pmatrix}, b = \begin{pmatrix} 4 \\ 2 \\ 8 \\ 0 \end{pmatrix}, c = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 12 \end{pmatrix} \Rightarrow \begin{cases} a = 0.5b + 0.25c \\ c = 4a - 2b \end{cases}$$

- otherwise they are **linearly independent**

recall on linear algebra

MEASURE OF
↑
SIMILARITY

- the **inner product** of 2 vectors $a = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$, $b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$ can be computed as



$$a^T b = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = \|a\| \|b\| \cos \alpha \quad (\alpha \text{ angle between } a \text{ and } b)$$

$$a = \begin{pmatrix} 2 \\ 1 \\ 4 \\ 3 \end{pmatrix}, b = \begin{pmatrix} 3 \\ 0 \\ 1 \\ 2 \end{pmatrix}, a^T b = (2 \ 1 \ 4 \ 3) \times \begin{pmatrix} 3 \\ 0 \\ 1 \\ 2 \end{pmatrix} = 2 \times 3 + 1 \times 0 + 4 \times 1 + 3 \times 2 = 16$$

- 2 vectors are **orthogonal** if their inner product is zero (they are perpendicular)

$$a = \begin{pmatrix} 2 \\ 0 \\ -1 \\ 3 \end{pmatrix}, b = \begin{pmatrix} 0 \\ 6 \\ 3 \\ 1 \end{pmatrix}, a^T b = (2 \ 0 \ -1 \ 3) \times \begin{pmatrix} 0 \\ 6 \\ 3 \\ 1 \end{pmatrix} =$$

$$2 \times 0 + 0 \times 6 - 1 \times 3 + 3 \times 1 = 0$$

THEY ARE
COMPLETELY
DIFFERENT



recall on linear algebra

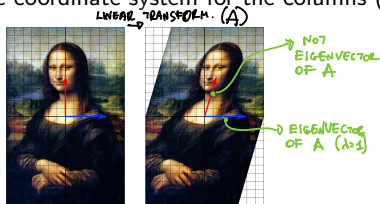
- given a matrix $A^{m \times n}$, its **rank** $rk(A)$ is the maximum number of columns or rows linearly independent ($rk(A) \leq \min\{m, n\}$) $rk(A) \rightarrow$ DIMENSION OF THE SPACE GENERATED BY THE COLUMNS (ROWS) OF A
- given a square matrix $A^{n \times n}$, a scalar λ and a vector $v \in \mathbb{R}^n$ are an **eigenvalue** and an **eigenvector** if they satisfy

$$Av = \lambda v$$

- eigenvectors of A are linearly independent
- if A is symmetric there are n orthogonal eigenvectors (associated to n REAL eigenvalues)
- provide an alternative coordinate system for the columns (rows) of A

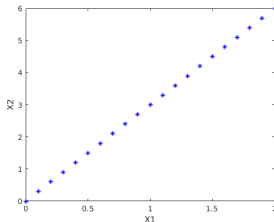
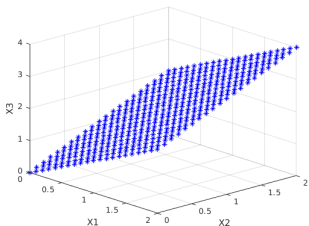


MATRIX $A \equiv$ LINEAR TRANSFORM.
 $x \rightarrow Ax$ LIN. TRANSFORM.
 1) ROTATION OF x
 2) MAGNIFICATION



why linear independence, orthogonality, matrix rank, eigenvalues/eigenvectors?
vectors are measures of variables $\Rightarrow$ investigating vectors relations provide insights on data (dependence, redundancy,...)

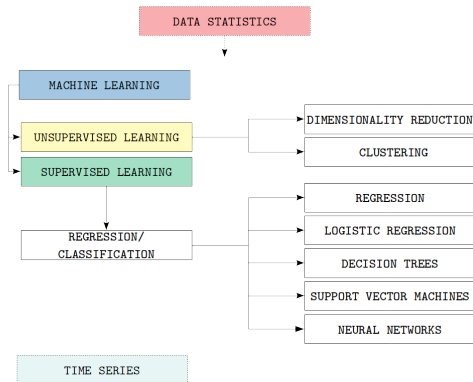
- a set of points with n coordinates may lie in a space of dimension $\leq n$
(points in a plane can lie in a line, points in a 3d-space can lie in a plane...)



- given a set of n dimensional points one may wonder what is the space of **minimum dimension** where they lie
- in such a space their mutual distance would be preserved
- maybe we are happy with something "less precise", ignoring the less relevant dimensions!



Principal Component Analysis (PCA)



PCA is a **dimensionality reduction** technique used for

- **exploratory data analysis** (unsupervised learning)
- **feature extraction** (support to supervised learning)